

Planetary Science & Solar System Dynamics

Problem Set

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This problem set accompanies the *Planetary Science & Solar System Dynamics* handout. Section A has five objective (multiple-choice) questions; Section B has three subjective questions, pitched a little harder than Section A. Use $S_{\odot} = 1361 \text{ W/m}^2$ at 1 AU, $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$, $GM_{\odot} = 4\pi^2 \text{ AU}^3/\text{yr}^2$ (so $1 \text{ AU/yr} = 4.74 \text{ km/s}$), $R_{\odot} = 6.96 \times 10^8 \text{ m}$, and $1 \text{ AU} = 1.496 \times 10^{11} \text{ m}$ where needed, unless a question tells you otherwise.

Section A — Objective

O1. An airless planet orbits the Sun at $d = 2 \text{ AU}$ with Bond albedo $A = 0.30$. Its effective temperature is closest to:

- (a) 150 K
- (b) 180 K
- (c) 220 K
- (d) 255 K

O2. A planet with effective temperature $T_{\text{eff}} = 255 \text{ K}$ has an atmosphere modelled as $n = 3$ ideal greenhouse layers. Its surface temperature is closest to:

- (a) 302 K
- (b) 360 K
- (c) 510 K
- (d) 1020 K

O3. Earth ($P_1 = 1.00 \text{ yr}$) and Mars ($P_2 = 1.88 \text{ yr}$) have a synodic period closest to:

- (a) 0.88 yr
- (b) 1.88 yr
- (c) 2.14 yr
- (d) 2.88 yr

O4. Which one of the following statements is **false**?

- (a) Quadrature is defined only for superior planets.
- (b) A superior planet is at its closest and brightest at opposition.
- (c) A body's geometric albedo can exceed 1, while its Bond albedo cannot.
- (d) An inferior planet reaches opposition once per synodic period.

O5. Taking the Moon's radius as $R_p = 1.737 \times 10^6$ m and its distance from the Sun as 1 AU, the length of the Moon's umbral cone is closest to:

- (a) 1.4×10^5 km
- (b) 3.7×10^5 km
- (c) 7.5×10^5 km
- (d) 3.7×10^6 km

Section B — Subjective

S1. A spacecraft is to be sent from Earth ($r_1 = 1.000$ AU, $P_1 = 1.000$ yr) to Mars ($r_2 = 1.524$ AU, $P_2 = 1.881$ yr) on a Hohmann transfer. Treat both orbits as circular and coplanar.

- (i) Find the semi-major axis a_t of the transfer ellipse, and hence the transfer time t_{transfer} in days.
- (ii) Using the vis-viva equation, find Δv_1 (departure burn), Δv_2 (arrival burn), and the total mission cost Δv_{total} , in km/s.
- (iii) Find the launch phase angle γ_0 that Mars must have at departure, and the synodic period S — i.e. how long the mission planners must wait between successive launch windows.

S2. A 1000 kg spacecraft performs a flyby of Jupiter ($GM_p = 1.267 \times 10^{17}$ m³/s², $M_p = 1.898 \times 10^{27}$ kg) with hyperbolic excess velocity $v_\infty = 10.0$ km/s and impact parameter $b = 1.00 \times 10^9$ m.

- (i) Find the turning angle δ of the flyby.
- (ii) Find the maximum possible heliocentric speed boost Δv_{helio} this encounter could deliver, and comment on how it compares to v_∞ itself.
- (iii) The boost in (ii) is not free — it is taken from Jupiter's own orbital motion. Using momentum conservation, estimate the magnitude of the resulting change in Jupiter's orbital speed, and comment on your answer.

S3. Venus orbits at $d = 0.723$ AU with Bond albedo $A = 0.77$. Its observed mean surface temperature is $T_s \approx 737$ K.

- (i) Find the solar flux S at Venus's orbit, and hence its effective temperature T_{eff} .

- (ii) Using the n -layer greenhouse model, $T_s^4 = (n + 1)T_{\text{eff}}^4$, find the number of layers n needed to reproduce the observed surface temperature.
- (iii) Venus receives roughly twice the solar flux Earth does, yet your answer to (i) should come out *colder* than Earth's effective temperature of about 255 K. Explain why, and then comment briefly on what your value of n in (ii) does and does not tell you physically — in particular, whether the model's layers should be read as real, countable atmospheric strata.